RESNET

How Skip Connections Changed Deep Learning

A critical case study of residual learning in convolutional neural networks

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1. Introduction:

1.1 Background:

During early 2010s, deep learning became a major breakthrough in computer vision, particularly in image classification and object recognition. Convolutional Neural Networks (CNNs) achieved impressive results by automatically learning features from images, progressing from single edges and textures to more complex shapes and objects. The success of AlexNet in the 2012 ImageNet Large Scale Visual Recognition Challenge (ILSVRC) demonstrated that deep CNNs could significantly outperform many traditional image-classification methods.

Following this success, researchers began developing deeper neural networks to improve recognition accuracy. The underlying idea was that increasing network depth would allow a model to learn more complex and abstract representations of visual information. Architectures such as VGGNet and GoogLeNet further demonstrated the potential of deep neural networks. However, as researches attempted to build increasingly deep models, an important limitation became apparent: **simply adding layers did not always improve performance and could sometimes make the network harder to train.**

1.2 Why was ResNet Proposed?

Before ResNet, increasing the depth of a CNN was considered a promising way to improve its ability to represent complex patterns. However, researchers observed that very deep networks could experience a problem known as the degradation problem. As additional layers were added, the training accuracy of a plain network could become worse, even when the network was not overfitting.

This behaviour was unexpected because a deeper network should theoretically be able to perform at least as well as a shallower one. In principle, the extra layers could learn an identity mapping, allowing the deeper network to preserve the original network's performance. In practice, however, conventional networks often struggled to learn such mappings through standard training. The problem was not simply that deeper networks required more computation or had more parameters. Rather, optimising very deep networks became increasingly difficult. Gradients had to propagate through many layers, and the additional layers did not necessarily learn useful transformations. Consequently, deeper models could achieve higher training error than their shallower counterparts.

To address this limitation, Kaiming He and his colleagues introduced the Residual Network (ResNet) architecture in their 2015 paper, Deep Residual Learning for Image Recognition, which was published at the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) in 2016. Their central idea was to make the learning process easier by allowing layers to learn residual functions instead of directly learning the complete desired mapping.

Although VGGNet and GoogLeNet achieved excellent results, they did not fully resolve the difficulty of optimising increasingly deep conventional networks. The leading ImageNet architectures at the time generally contained approximately 16–30 layers. ResNet challenged this limit by demonstrating that networks with substantially greater depth could be trained effectively.

1.3 Residual Learning Approach:

To overcome the degradation problem, Kaiming and his colleagues proposed that a group of layers should learn a residual function rather than directly learn the complete desired mapping. In a conventional network, a group of layers attempts to learn a mapping H(x), where x represents the input. In a residual block, the layers instead learn the difference between the desired mapping and the original input:

$$F(x) = H(x) - x$$

The output is then expressed as:

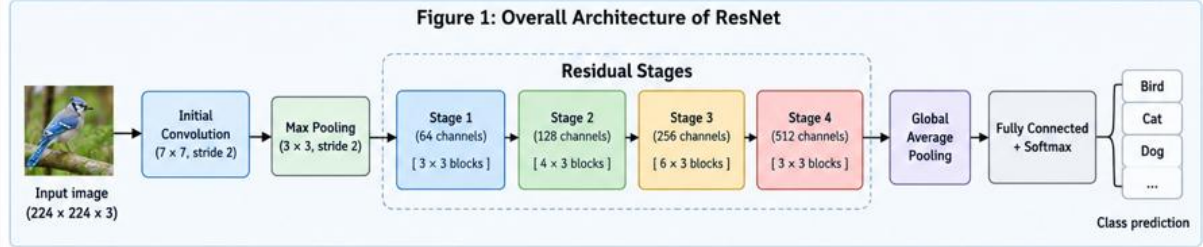
$$H(x) = F(x) + x$$

Here, F(x) represents the residual transformation, while \(x\) is passed directly to the output through a shortcut connection. This allows the network to preserve the input information and learn only the necessary modifications. If no modification is needed, the residual function can ideally approach zero, making identity mapping easier to represent.

2. Architecture:

2.1 Overall Architecture:

The network begins with a convolutional layer that extracts basic features such as edges and textures. The input then passes through several residual blocks, where these features are gradually developed into more complex representations. As the network becomes deeper, the spatial dimensions of feature maps decrease while the number of channels increases. Finally, global average pooling and a fully connected layer are used to predict the image category.

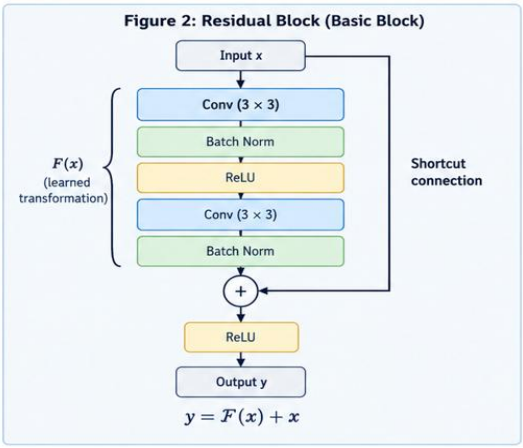


2.2 Residual Blocks and Shortcut Connections:

The most important design feature of ResNet is its residual block. In a traditional CNN, each group of layers learns a complete transformation of its input. ResNet instead learns a residual function, which represents the change required in the input features.

The relation is expressed as: [y=F(x)+x]

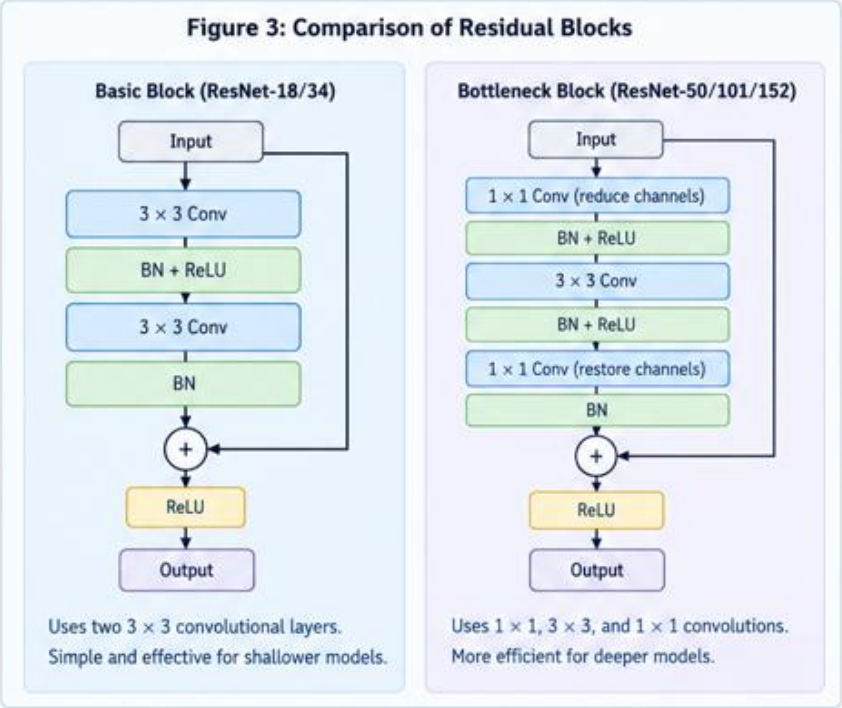
Here, (x) is the input, (F(x)) is the learned transformation, and (y) is the output. The shortcut connection carries the original input directly to the addition operation. This design was introduced deeper networks often struggled to learn identity mappings. By allowing the input to bypass some layers, ResNet makes information and gradients easier to propagate through the network. As a result, much deeper models can be trained more effectively.



2.3 Basic Blocks and Bottleneck Blocks:

ResNet uses two main types of residual blocks. The basic block contains two (3 \times 3) convolutional layers and is used in ResNet-18 and ResNet-34. Deeper models, such as ResNet-50, ResNet-101 and ResNet-152, use bottleneck blocks consisting of (1 x 1), (3 x 3) and (1 x 1) convolutions.

The bottleneck design reduces the number of channels before the main convolution and restores them afterwards. This helps control computational cost while allowing the network to become deeper. When input and output dimensions differ, a projection shortcut is used to make their dimensions compatible.



3. Strengths and Limitations:

3.1 Strengths:

One of the main strengths of ResNet is its ability to train very deep networks using residual connections. These connections help information and gradients pass through the network, reducing the optimisation difficulties associated with conventional deep CNNs. This allows ResNet to learn complex visual patterns and achieve strong predictive performance in image classification tasks.

Another advantage is its flexibility. Different versions, such as ResNet-18, ResNet-50 and ResNet-152, provide options for different computational requirements. Smaller models are suitable when speed and efficiency are important, while deeper models can capture more complex features when sufficient resources are available.

ResNet is also useful beyond image classification. It is widely used as a backbone for object detection, image segmentation, medical image analysis and industrial defect detection. For ex, in medical imaging, ResNet-based models can help identify patterns in X-rays and other scans. However, their predictions still require appropriate clinical validation before they can support real-world medication decisions.

3.2 Limitations:

Despite its advantages, ResNet can require considerable computational power, particularly in its deeper variants. Training large models may need powerful GPUs, substantial memory and more time. Therefore, deploying a large ResNet model on mobile devices or other resource-constrained systems can be challenging.

Increasing the depth of the network also increases its computational complexity, and it does not guarantee better performance in every situation. A deeper model may provide limited improvement when the dataset is small or the task is relatively simple. It may also overfit if the training data are insufficient or unrepresentative.

Another limitation is that ResNet primarily learns visual features from the data provided to it. In applications such as medical imaging, biased datasets, poor image quality or differences between hospitals can affect its generalisation to new cases. Furthermore, its predictions can be difficult to interpret, which is an important concern in applications where decisions must be explained.

3.3Trade-offs and Practical Applications:

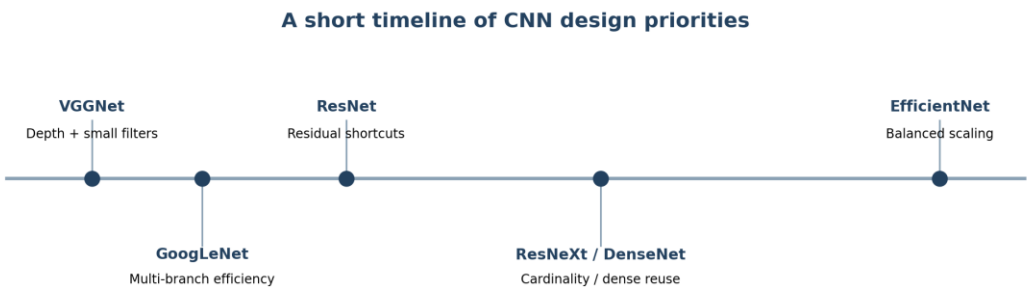
The choice of ResNet variant involves a trade-off between predictive performance and computational cost. ResNet-18 is generally faster and lighter, making it useful for applications where processing speed matters. ResNet-50 offers a deeper bottleneck architecture that can provide stronger feature representations, but it requires more computation. ResNet-152 can capture highly complex patterns, although its additional cost may not be justified for every task.

In medical imaging and industrial inspection, higher predictive accuracy may be valuable because missed abnormalities or defects can have serious consequences. However, these applications also require careful validation, reliable data and suitable hardware. In real-time applications, a smaller model may be preferable if it provides sufficient accuracy with lower latency.

Overall, ResNet offers a strong balance between predictive performance and trainability, but the best model depends on the specific problem. Its depth, computational requirements, generalisation and interpretability must all be considered before choosing it for practical use.

4. Subsequent Developments:

4.1 Evolution of CNN



Later work retained some ResNet ideas while exploring other ways to improve networks. ResNeXt kept residual connections but aggregated multiple transformations, making cardinality—the number of parallel transformations—another design dimension (Xie et al., 2017). DenseNet connected each layer to many later layers to encourage feature reuse and improve information flow (Huang et al., 2017). These designs differ in how features travel between layers, but share the insight that connectivity matters.

4.2 What Lasted and What Changed?

The evolution of CNNs shows that researchers gradually moved beyond simply increasing network depth. They began focusing more on feature reuse, computational efficiency, balanced model scaling and the ability to learn relationships across images.

ResNet's shortcut connections remained influential in later architectures because they help information flow through deep networks. Convolutional layers also continued to play an important role in image processing. However, the exact block structures and methods of connecting layers evolved to suit different tasks and computational requirements.

Overall, later architectures did not make ResNet irrelevant. Instead, they extended its central ideas and introduced alternatives for specific needs. ResNet remains a useful baseline, while models such as MobileNet and EfficientNet are attractive when efficiency matters, and architectures such as ViT offer a different approach to learning visual features

5. Critical Assessment:

ResNet addressed a specific and important problem in CNN development: deeper plain networks could become harder to optimise rather than automatically becoming better. By learning residual functions and adding shortcut connections, it offered a simple, repeatable way to train substantially deeper models. Its success in ImageNet and other visual recognition tasks helped establish residual learning as a standard design principle.

However, ResNet should not be treated as a universal solution. Model size, compute requirements, domain shift and dataset quality continue to limit performance in real applications. Later architectures have improved efficiency, changed connectivity patterns and explored alternatives to convolution, but many still retain residual connections. Overall, ResNet’s lasting contribution is the principle that successful deep architectures depend not only on how many layers they contain, but also on how information and gradients move through them. That principle remains relevant when designing and evaluating contemporary neural networks.

The lasting lesson of ResNet is not simply that more layers are better. It is that the route by which information moves through a network can be as important as the operations performed by individual layers. Residual connections made very deep CNNs more practical and became

a reusable building block in later vision systems and other deep-learning architectures. At the same time, the continued development of efficient CNNs shows that depth must be balanced against computational cost. ResNet remains important both as a practical baseline and as a turning point in the way researchers think about training deep networks.

References

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